

Straight Lines

Week 4B

Overview

Properties of Straight Lines and Equations of Lines

Key Ideas - Properties of Linear Graphs

- Equations of Straight Lines

Key Idea – Properties of Linear Graphs

Most linear equations can be written in the form $y = mx + b$, where m and b are constants.

- The constant m is the coefficient of x and represents the **gradient** of the graph of the equation.
- The constant term b is the **y-intercept**, which is the point where the line cuts the y -axis.

The Intercepts of a Straight Line

Most straight lines have two intercepts.

The **y-intercept** is where the graph cuts the y -axis, i.e. when $x = 0$. The symbol for the y -intercept is b , as mentioned above.

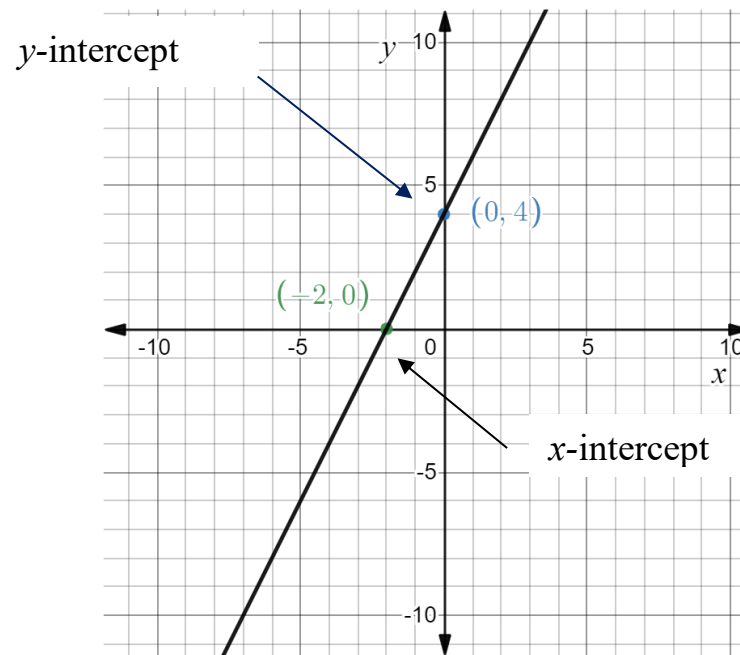
The **x-intercept** is where the graph cuts the x -axis, i.e. when $y = 0$. There is no symbol for the x -intercept, which is more difficult to obtain from the equation.

Example:

The straight line drawn below:

Cuts the y -axis at 4, so the coordinates of the y -intercept are $(0, 4)$. This can also be written as $b = 4$.

Cuts the x -axis at -2 , so the coordinates of the x -intercept are $(-2, 0)$.



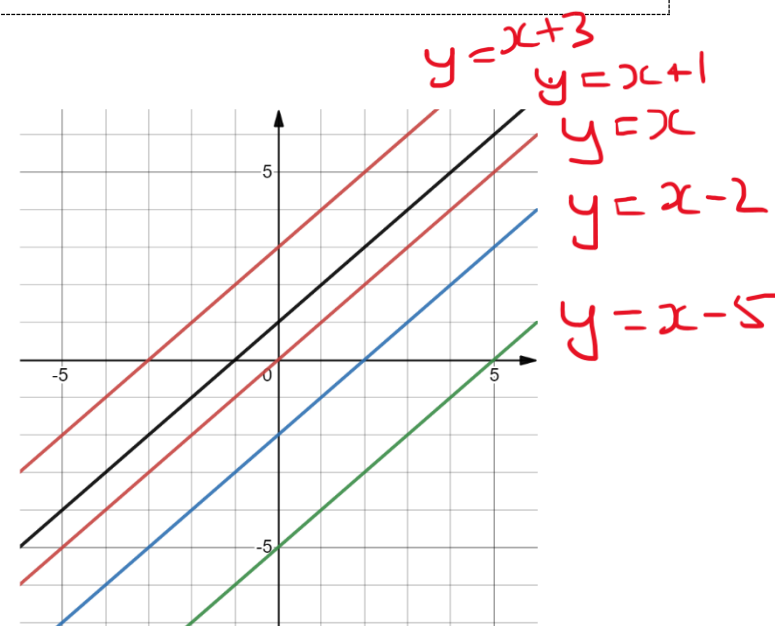
Discussion Question:

By using the graphing package “Desmos”, we will investigate the graphs of the following functions: $y = x$, $y = x + 1$, $y = x + 3$, $y = x - 2$, and $y = x - 5$.

Comment on the effect of changing the value of the constant term.

It moves the line up & down

From these graphs we note that, as the **constant term** is changing, the **y-intercept** of the graph changes. The constant term determines the **y-intercept** of the line.

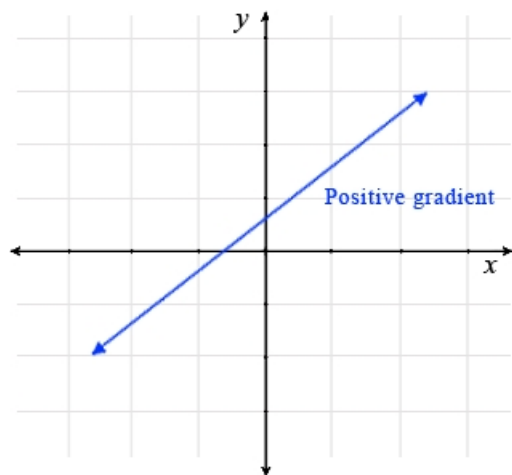


The Gradient of a Straight Line

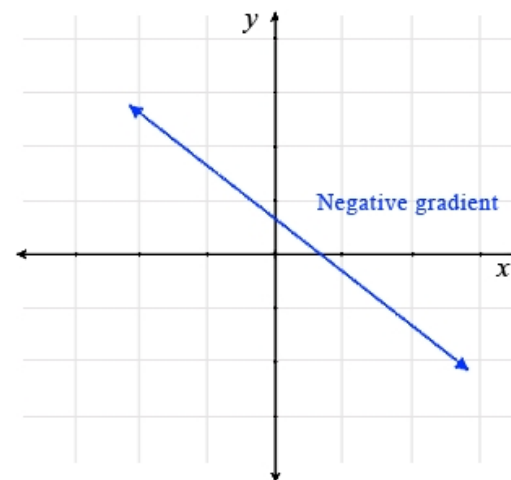
The **gradient** or **slope** of a line is a numerical value that describes its steepness and the direction in which it points. The gradient is denoted by the letter ***m***.

Direction of a Line

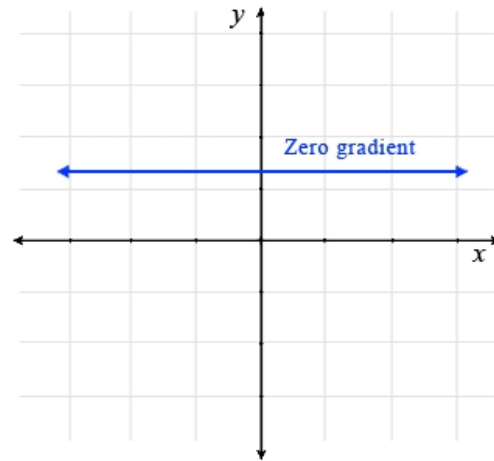
A **positive value of m** ($m > 0$) describes a graph which slopes **upwards** from left to right.



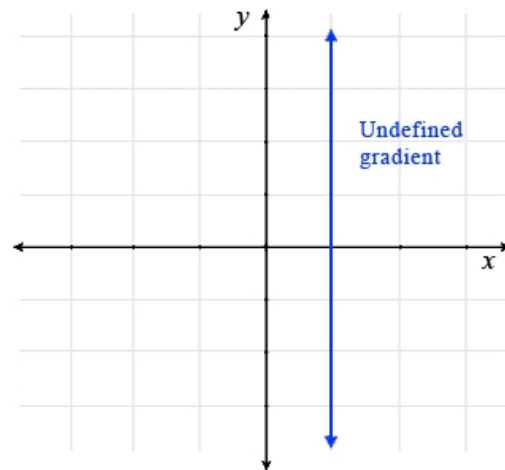
A **negative value of m** ($m < 0$) describes a graph which slopes **downwards** from left to right.



The **gradient** of a **horizontal** line is **zero**.



The **gradient** of a **vertical** line is **undefined**.



Discussion Question:

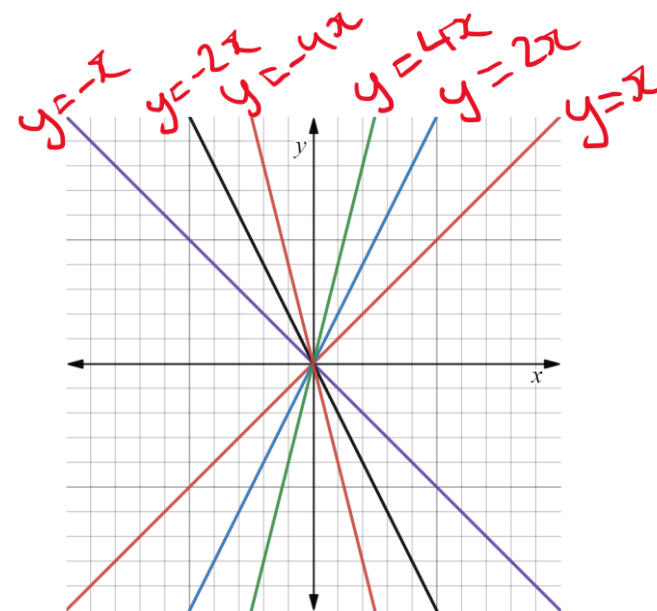
By using the graphing package “Desmos”, we will investigate the graphs of the following functions: $y = x$, $y = 2x$, $y = 4x$, $y = -x$ and $y = -2x$.

Comment on the effect of changing the value of the coefficient of x .

it changes the slope of the line

From these graphs we note that, as the **coefficient of x** is changing, the **gradient of the graph changes**. The coefficient of x gives the **gradient** or **slope** of the line.

We will now look at how to determine the gradient of a line.



Magnitude of the Gradient

The gradient or slope of a line **determines how steep the line is and in which direction it points**. It is denoted by the variable ***m***.

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two point on the plane. The gradient of the interval from AB is measure of its steepness. The gradient of the interval AB is defined to be the ratio of the rise and the run.

The gradient of the line joining $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

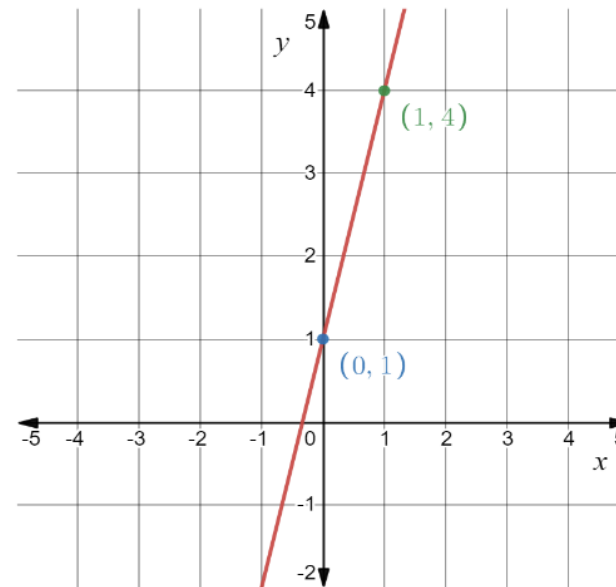
$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example:

Determine the slope of the following line.

Pick any two points on the line, e.g. (0, 1) and (1, 4). Then substitute values into the gradient formula:

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{1 - 0} \\ &= 3 \end{aligned}$$



Discussion Question:

What can you **conclude** if we use the same line as given above but use two different points.

$$\text{lets use } (x_1, y_1) = (-1, -2) \\ (x_2, y_2) = (1.5, 5.5)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5.5 - (-2)}{1.5 - (-1)} = \frac{7.5}{2.5} = 3$$

The gradient is the same

Note:

Although you can pick 'any' two points, ensure that the coordinate values of the two points are integer values (whole numbers) to avoid estimation errors.

Discussion Question:

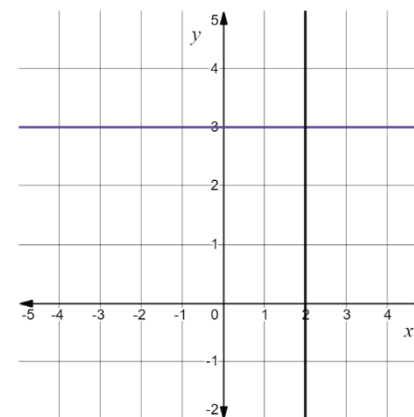
Find the gradient of the horizontal and vertical lines

- for horizontal line: $(x_1, y_1) = (0, 3)$
 $(x_2, y_2) = (1, 3)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 3}{1 - 0} = \frac{0}{1} = 0$$

- for vertical line: $(x_1, y_1) = (2, 0)$
 $(x_2, y_2) = (2, 1)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{2 - 2} = \frac{1}{0} = \text{undefined}$$



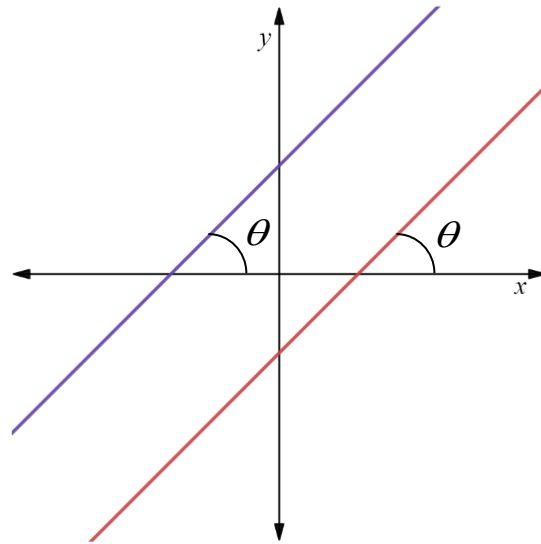
Note:

Vertical lines don't have a gradient – the run is always zero, so the fraction is undefined.

Horizontal lines have gradient zero, because the rise is always zero.

Parallel Lines

Parallel lines make angles of equal magnitude with the positive direction of the x -axis, i.e. their angle of inclination is the same, i.e they have the **same gradient**.



Two lines are parallel if their gradients are equal,

$$m_1 = m_2$$

Example: Show that the line $y = 3x + 2$ is parallel to the line $2y - 6x + 8 = 0$.

The line $y = 3x + 2$ is in the form $y = mx + b$, so the gradient $m = 3$

The line $2y - 6x + 8 = 0$ can be rearranged,

$$2y - 6x + 8 = 0$$

$$2y = 6x - 8 \quad \text{the gradient is also } m = 3$$

$$y = 3x - 4$$

Therefore the two lines are parallel as the gradients are the same.

Exercise:

Show that the line $2y - 4x + 2 = 0$ is parallel to the line $y = 2x + 4$

for line 1:

$$2y - 4x + 2 = 0$$

$$2y = 4x - 2$$

$$y = 2x - 1$$

$$\boxed{m_1 = 2}$$

for line 2:

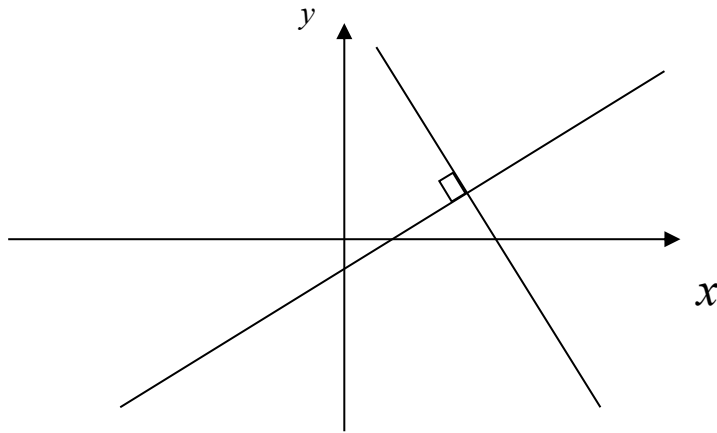
$$y = 2x + 4$$

$$\boxed{m_2 = 2}$$

$m_1 = m_2 \rightarrow$ the lines are parallel

Perpendicular lines

Perpendicular lines **intersect at 90°** .



If the slope of one line is m_1 , and the slope of a perpendicular line is m_2 , then **$m_1 \times m_2 = -1$** .

Example:

If a line has a slope of 4, then a perpendicular line will have slope of $-\frac{1}{4}$.

Exercise:

(a) Show that the two lines $3x - 4y = -5$ and $4x + 3y - 8 = 0$ are perpendicular.

for line 1:

$$3x - 4y = -5$$

$$3x + 5 = 4y$$

$$4y = 3x + 5$$

$$y = \frac{3}{4}x + \frac{5}{4}$$

$$\boxed{m_1 = \frac{3}{4}}$$

for line 2

$$4x + 3y - 8 = 0$$

$$3y = -4x + 8$$

$$y = -\frac{4}{3}x + \frac{8}{3}$$

$$\boxed{m_2 = -\frac{4}{3}}$$

$$m_1 \times m_2 = \frac{3}{4} \times \left(-\frac{4}{3}\right)$$

$$= -1$$

$\therefore$ lines are perpendicular

(b) Show that the line joining (1, 2) and (3, 4) is perpendicular to the line $y = -x + 3$.

for line 1:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
$$= \frac{4 - 2}{3 - 1}$$

$$\boxed{m_1 = 1}$$

for line 2:

$$y = -x + 3$$

$$\boxed{m = -1}$$

$$m_1 \times m_2 = 1 \times (-1)$$
$$= -1$$

$\therefore$ lines are perpendicular

Key Idea – Equations of Straight Lines

Gradient Intercept Form

Every linear equation can be expressed in the form

$$y = mx + b \quad \text{where } m \text{ and } b \text{ are constants.}$$

The constant ***m*** is the coefficient of *x* and is referred to as the **gradient** of the graph of the equation and ***b*** is the constant term and called the ***y* – intercept**.

This form is said to be in **gradient-intercept form**.

Note:

*If the equation to the straight line is **not** in the form $y = mx + b$, it is necessary to re-arrange it, so that the subject of the equation is *y* when finding the gradient and the *y* – intercept.*

Example: Determine the gradient and the y-intercept of the line with equation $y = 3x + 1$.

$$m = 3 \text{ and } b = 1$$

Exercise:

Determine the gradient and the y-intercept of the line with the equation $5y + 2x = -15$.

$$5y = -2x - 15$$

$$y = -\frac{2}{5}x - \frac{15}{5}$$

$$y = -\frac{2}{5}x - 3$$

- gradient = $-\frac{2}{5}$
- y-intercept = -3

General Form

A linear equation can also be written in **general form**. The general form of the equation of a straight line is:

$$ax + by + c = 0$$

where a , b and c are integers and $a > 0$.

Example: The following are **not** general forms of a linear equation:

$$-3x + 8y - 3 = 0 \quad \text{as } a < 0$$

$$x + y - 0.5 = 0 \quad \text{as } c = -0.5 \text{ is not an integer.}$$

Exercise: Write the following linear equation in general form $y = -\frac{5}{2}x + \frac{9}{5}$.

$$10 \times y = 10 \left(-\frac{5}{2}x + \frac{9}{5} \right)$$

$$10y = -25x + 18$$

$$25x + 10y - 18 = 0$$

Exercise:

Is the equation $3x + y - 5 = 0$ the same as $y = 5 - 3x$?

rewrite $3x + y - 5 = 0$

$$y = -3x + 5$$

$$= 5 - 3x$$

Yes they are the same.

Point Gradient Form

The straight line with slope m , and passing through the point (x_1, y_1) has equation

$$y - y_1 = m(x - x_1)$$

Example: Find the equation of the straight line with a slope -3 which passes through point $(1, 2)$

We have the slope, $m = -3$ and the point $(x_1, y_1) = (1, 2)$

Now we need to substitute into the above equation.

$$y - 2 = -3(x - 1)$$

$$\Rightarrow y = -3x + 3 + 2$$

$$\Rightarrow y = -3x + 5 \quad \text{gradient intercept form}$$

$$\text{or } 3x + y - 5 = 0 \quad \text{general form}$$

Exercise:

Find the equation of the straight line with a slope of $-\frac{3}{4}$ which passes through point $(1, -4)$.

Leave your answer in general form.

use

$$y - y_1 = m(x - x_1)$$

$$y - (-4) = -\frac{3}{4}(x - 1)$$

$$y + 4 = -\frac{3}{4}x + \frac{3}{4} \quad (-4)$$

$$y = -\frac{3}{4}x - \frac{13}{4} \quad (\times 4)$$

$$4y = 4\left(-\frac{3}{4}x - \frac{13}{4}\right)$$

$$4y = -3x - 13$$

$$3x + 4y + 13 = 0$$

Two Point Formula

The straight line passing through the **points** (x_1, y_1) and (x_2, y_2) has equation

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Note: Alternatively, you can find the **slope** m between the two points (x_1, y_1) , (x_2, y_2) . Then use the point gradient formula using coordinates of either of the two points.

Example: Find the equation of the straight line passing through the two points $(-1, 3)$ and $(3, 8)$ and write it in general form.

We need to use $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$

$$\frac{y - 3}{x - (-1)} = \frac{8 - 3}{3 - (-1)}$$

$$\frac{y - 3}{x - (-1)} = \frac{5}{4}$$

$$4(y - 3) = 5(x + 1)$$

$$4y - 12 = 5x + 5$$

$$5x - 4y + 17 = 0$$

Exercise: P and Q are the points $(2, -3)$ and $(0, 1)$, find the equation of the straight line PQ .

using

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\frac{y-(-3)}{x-2} = \frac{1-(-3)}{0-2}$$

$$\frac{y+3}{x-2} = \frac{4}{-2}$$

$$y+3 = -2(x-2)$$

$$y+3 = -2x+4$$

$$y = -2x+1$$

$$2x+y-1=0$$

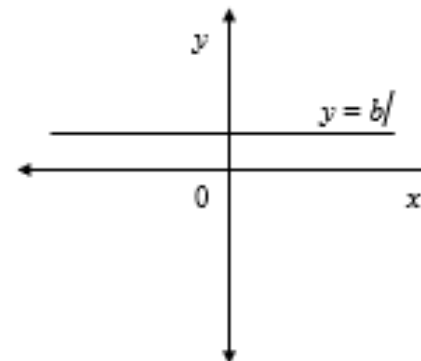
(in gradient intercept form)

(in general form)

Lines Parallel to Axes

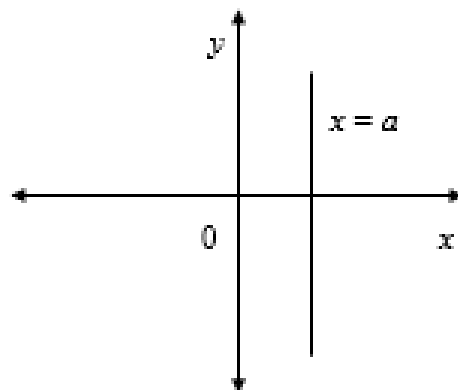
Horizontal Lines

If a line is parallel to the x -axis, it has a gradient of zero i.e. $m = 0$ and cuts the y -axis at $(0, b)$, then the equation is $y = b$.



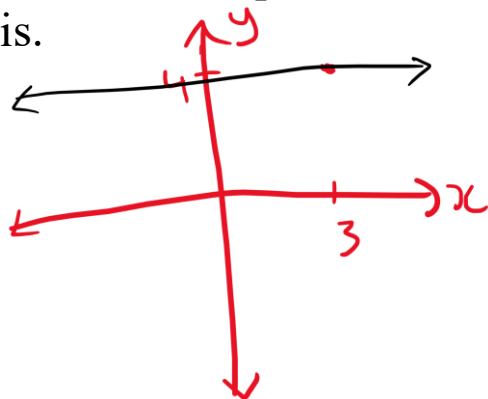
Vertical Lines

If a line is parallel to the y -axis, it has **no gradient** and cuts the x -axis at $(a, 0)$, then the equation is $x = a$.



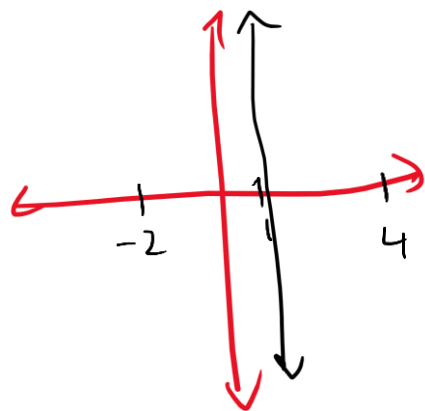
Exercise:

- (a) Find the equation of the line which passes through the point (3, 4) and is parallel to the x-axis.



equation of the line: $y=4$

- (b) Find the equation of the line which is parallel to the y-axis and midway between the lines $x = -2$ and $x = 4$.



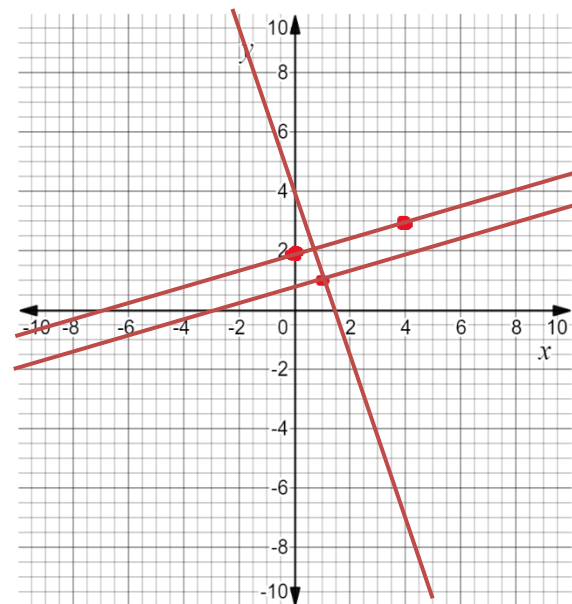
$$\begin{aligned} \text{halfway between } x = -2 \text{ \& } x = 4 \\ &= \frac{-2 + 4}{2} \\ &= 1 \end{aligned}$$

equation of the line: $x=1$

Finding the Equation of Parallel and Perpendicular Lines

Discussion Question:

Draw a line that passes through the points $(0,2)$ and $(4,3)$.



Now draw a line that passes through $(1,1)$ and is parallel to the line drawn above.

Now draw a line that passes through $(1,1)$ and is perpendicular the line drawn above.

Exercise:

Find the equation of the line which passes through the point (1, 4) and is parallel to the line $3x - y + 8 = 0$. Leave your answer in gradient-intercept form.

- first find the gradient of the line

$$3x - y + 8 = 0$$

$$y = 3x + 8$$

$$\boxed{m = 3}$$

$$(x_1, y_1) = (1, 4)$$

$$m = 3$$

- since the line is parallel use $m = 3$

- now use $y - y_1 = m(x - x_1)$

$$y - 4 = 3(x - 1)$$

$$y - 4 = 3x - 3$$

$$\boxed{y = 3x + 1}$$

Exercise:

Find the equation of the line perpendicular to $x - 2y = 6$ which passes through $(1, 3)$. Give your answer in general form.

- first find the gradient of the line

$$x - 2y = 6$$

$$2y = x - 6$$

$$y = \frac{1}{2}x - \frac{6}{2}$$

$$\boxed{m_1 = \frac{1}{2}}$$

$$(x_1, y_1) = (1, 3)$$

$$m = -2$$

- find m_2 using

$$m_2 = -\frac{1}{m_1} = -\frac{1}{\frac{1}{2}} = \boxed{-2}$$

- now use

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -2(x - 1)$$

$$y - 3 = -2x + 2$$

$$\boxed{2x + y - 5 = 0}$$